Experiment 2, Part A: Golden Gate Assembly and Transformation of Binders, Targets and Reporter

Purpose

Clone each computationally designed MEGF8 and MOSMO binder into the activation-domain vector by SapI Golden Gate Assembly, then transform E. coli to recover the assembled plasmids for the split-transcription-factor binding screen (Experiment 2). The MEGF8 and MOSMO targets are cloned into the GAL4 DNA-binding-domain vector in the same way, and the reporter plasmid is transformed and grown for purification.

Materials and equipment

- Binder DNA plates supplied by Twist Bioscience
- Target DNA fragments by Twist Bioscience
- Activation-domain vector
- GAL4 DNA-binding-domain vector
- Reporter plasmid
- SapI (not BsaI), T4 DNA ligase and 10× T4 ligase buffer
- Nuclease-free water
- Echo acoustic liquid handler, Echo-compatible plates and the Echo transfer map
- Plate shaker and plate centrifuge
- 37 °C incubator and shaking incubator
- Chemically competent DH5α E. coli, SOC medium, repeater pipette
- LB with carbenicillin (1:1,000 dilution of stock), 96-well round-bottom deep-well plates and 50 mL culture vessels

Golden Gate reaction composition (2 µL per reaction)

Component	Per reaction (µL)	Notes
Water	1.381	
10× T4 ligase buffer	0.200	1× final
Vector	0.049	
SapI	0.120	1.2 U
T4 DNA ligase	0.100	40 U
Master mix subtotal	1.850	Prepare in proportion to the number of reactions
Binder DNA	0.150	From the binder plate
Total	2.000	

Procedure: Golden Gate Assembly

1. Prepare the reaction calculation for the number of wells, using the SapI calculator.
2. Spin down the binder plate(s) at 4,000 ×g for 1 minute.
3. Prepare the master mix from the table above.
4. Combine the master mix and the binder DNA on the Echo, following the Echo transfer map. Re-run any wells listed in the Echo exceptions report.
5. Shake the plate on the plate shaker for 1 minute.
6. Incubate at 37 °C for 1 hour.

Procedure: 96-well transformation

1. Using a repeater pipette, add 8 µL chemically competent DH5α cells to each reaction well.
2. Incubate on ice for 30 minutes.
3. Heat shock for 30 seconds at 42 °C.
4. Return to ice for 5 minutes.
5. Add 95 µL SOC to each well.
6. Shake for 1 hour at 37 °C.
7. Transfer the entire reaction from each well into 1 mL LB with carbenicillin in a 96-well round-bottom deep-well plate.
8. Shake for 12–16 hours at 37 °C and 1,000 rpm.

Procedure: Golden Gate Assembly of targets

1. Prepare the master mix as in the table above, using the GAL4 DNA-binding-domain vector, at tube scale: 5.55 µL master mix and 0.45 µL target DNA per tube.
2. Incubate at 37 °C for 1 hour.

Procedure: transformation of targets and reporter

1. For each target assembly, add the reaction to 50 µL of chemically competent DH5α cells. For the reporter, add the plasmid to 50 µL of DH5α cells without an assembly step.
2. Incubate on ice for 30 minutes.
3. Heat shock for 30 seconds at 42 °C.
4. Return to ice for 5 minutes.
5. Add 200 µL SOC.
6. Shake for 1 hour at 37 °C.
7. Transfer each entire reaction into 50 mL of LB with carbenicillin (1:1,000 dilution of stock) and shake overnight at 37 °C.

Controls

No separate assembly or transformation controls are run.

Expected readout

Because the binder format is 96 wells, plasmid identity is not individually confirmed for each well. Hits from the screen are sequence-verified.

Next step

Purify plasmid DNA from the overnight cultures: binders by miniprep, and targets and reporter by midiprep (Part B).